

Exploring the Evolution of Dark Patterns and Manipulative Design on Instagram

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Abstract

Dark patterns or manipulative design patterns, design strategies aimed at deceiving or manipulating users into unintended actions, have increasingly become a point of concern in UI and UX design. A gap exists in the deceptive design literature on case studies of social networking services, particularly longitudinal investigations of manipulative design patterns. To address this gap, we employ an empirical design analysis methodology involving collecting video footage from 2010 to 2024 and analyzing it based on recent taxonomies and inductive coding. We contextualize the manipulative design patterns that have emerged on Instagram into key themes: (1) Provision of limited and illusory control to users in response to negative user sentiment, (2) adaptation in response to policy change, and (3) manipulative promotion of sub-apps competing with other platforms. The findings aim to inform designers, legislators, and researchers by illustrating evolving manipulative design techniques. Contributions include a novel approach for obtaining archival material about UI interface changes, an “evolution-based” thematic case study of Instagram’s manipulative design patterns over its history, and the identification of four new deceptive design patterns.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI; HCI theory, concepts and models; Empirical studies in interaction design; Interaction design theory, concepts and paradigms.**

Keywords

social networking services, social media, Instagram, manipulative design, deceptive design, interface design, dark patterns, digital well-being, ethical interfaces

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1 Introduction

The UX and UI design practices often referred to as “dark patterns”, “deceptive design patterns”, or “manipulative design patterns”¹ have increasingly become a subject of concern and research over the last decade and a half. In 2010, cognitive scientist and designer Harry Brignull coined the term “dark pattern” while launching a website exposing manipulative design practices [5]. He defined a dark pattern as “a user interface carefully crafted to trick users into doing things they might not otherwise do”, “carefully crafted with a solid understanding of human psychology”, and not with “the user’s interests in mind” [4]. In 2018, Gray and colleagues introduced a strategy-based taxonomy for these deceptive design patterns, building on Brignull’s foundational work [9]. Later, Mathur and colleagues classified patterns into diverse thematic categories and related them to normative perspectives, drawing on economics, ethics, law, philosophy, and psychology [20]. Gray and colleagues’ more recent ontology has since aimed to create a common language for scholars, practitioners, and legislators by clustering patterns from previous taxonomies [11]. Deceptive design patterns have been explored to varying extents in a variety of contexts, including e-commerce websites [19], GDPR-related cookie consent pop-ups [26], mobile games [1], IoT devices [16], and among different demographics, such as nationalities [13, 32], and children [28, 33] and teenagers [8, 30]. Other works have also highlighted their prevalence in mobile apps and users’ general lack of awareness of them [6], as well as the effects of specific patterns [34].

Much research on Deceptive design patterns has involved broad taxonomies that span multiple domains and social media platforms have often been included as data points. Overlapping with scholarly taxonomies to varying degrees, regulatory bodies have also begun to publish guidance on deceptive design patterns in recent years; for example, the 2022 guidelines on social media deceptive design patterns from the European Data Protection Board (EDPB) [7]. Numerous studies have also been conducted specifically on social networking services [10, 21, 23–25, 29]. Some studies have focused on manipulative design patterns across several social networking services, such as Mildner and colleagues’ 2023 study that systematically examined patterns on four major platforms [23]. Other research has focused on certain characteristics common to social media platforms, such as the study on attention capture design patterns from Roffarello and colleagues [25] or account deletion from

¹We cherish our collective responsibility to champion the principles of justice, diversity and the oneness of humankind and its implications for thoughtful decision-making about language. The ACM Diversity and Inclusion Council has included the term “dark pattern” in a list of potentially problematic terminology. While some works continue to use it in certain cases, others have adopted alternative terminology. Since we mostly use the terms interchangeably in this work, we choose to generally employ the alternatives of “deceptive design pattern” and “manipulative design pattern”.

Schaffner and colleagues [31]. Other studies have provided further granularity by studying patterns and nuances related to individual platforms, for instance, Lukoff and colleagues' 2021 analysis of YouTube's design features through the lens of user agency [18]. More platform-specific analyses could uncover unique patterns in ways that broad taxonomies may not capture.

Studies on manipulative design patterns are often constrained to a particular point in time due to data collection methods that often rely on screenshots or brief videos captured at the time of the study. This temporally-bound approach inhibits exploration of the history and evolution of deceptive design patterns. A notable exception is the 2021 study by Mildner and Savino, which used empirical design analysis to examine the evolution of privacy-related manipulative design patterns within Facebook [22]. However, the authors acknowledged limitations in sourcing sufficient screenshots to conduct a thorough analysis.

To date, we have found no studies that have focused on the historical evolution of deceptive design patterns on Instagram, one of the most prominent social media platforms. The main contributions of our work are (1) a novel method for obtaining archival material on UI interface changes, (2) an "evolution-based" thematic case study of Instagram's deceptive design patterns from 2010 to 2024, and (3) the identification of four new deceptive design patterns.

2 Empirical Design Study Methodology

Building on the empirical design analysis approach of Mildner and Savino [22], we reproduce and enhance their methodology by examining new techniques to document evolutions of widely used platforms. Specifically, we leverage the large volume of user-generated tutorials, demos, and other videos on YouTube to capture snapshots of Instagram's interface from its inception to the present. Based on our observations, we highlight the evolution of manipulative design patterns on Instagram over a decade and a half, offering a more continuous perspective on such design patterns.

Our empirical design analysis involved five steps: video search list compilation, title-based filtering, app footage filtering, UI change documentation, and thematic coding and organization. Figure 1 and Table 1 outline these stages of the methodology. In Step 1, we conducted manual Google custom searches to locate YouTube videos that showcased the Instagram app from 2010 to 2024. This involved a range of search terms to ensure sufficient sources covering various parts of the app. Our search terms used pairwise combinations of the keyword "Instagram" with each keyword from a set intended to capture general interface usage, specific features, and sub-apps: "tutorial", "review", "demo", "update", "how to", "privacy", "delete", "chronological", "explore", "setting", "settings", "feature", "features", "threads", "IGTV", "reels", "Meta AI". We conducted searches iteratively using custom date ranges from the first to last day of each year, from 2010 to 2024, to ensure sufficient footage from the platform's history. Approximately 7,800 video links were collected.

We used Google Apps Script integrated with the YouTube Data API to collect title and date metadata. In Steps 2 and 3, we manually filtered multiple times based on relevance. In Step 2, we reviewed and filtered video titles to exclude irrelevant videos, such as tutorials for non-Instagram software (including image and video editing software and apps, and programming tutorials), vlogs, lifestyle videos,

news segments, filmmaking videos, product reviews, interviews, and music-related content. This reduced the dataset to 1,081 videos. In Step 3, we skimmed each video to ensure it contained interface footage, further reducing the dataset to 821. We focused on the Instagram mobile app and excluded footage from desktop, Apple Watch, and other platforms. In Step 4, we chronologically (by date of publication) viewed each of the 821 videos to observe notable interface features and changes. We made a total of 228 notes on 155 of the videos watched, noting instances of interface changes in icons, buttons, tabs, features, sub-apps, settings, etc. and instances of manipulative design patterns distinct from previous videos.

In Step 5, we coded the data in three iterative rounds. In the first round, we inductively coded notes based on numerous aspects of the app. In the second round, we grouped these initial codes into higher-level categories. The higher-level categories and detailed aspects of the apps were Tabs: Home Feed, Explore Tab, Notifications Tab; Settings: Single-Setting Pages, Defaults, Account Deletion, Log Out, Miscellaneous Settings; Sub-Apps: Stories, IGTV, Reels, Threads (Snapchat competitor), Threads (X competitor), Meta AI; Account Creation and Profile: Policies, Account Creation and Profile, Miscellaneous; Other: People to Follow Suggestions, In-App Browser, Bottom Menu Bar, Top Menu Bar, Logged Out App Use, Mobile Browser Use, Access to Contacts and Facebook Friends. In the third round, we focused on a selection of categories that were most insightful to the evolution of manipulative design patterns and formulated three overarching themes.

We also used a few supplementary sources including web pages, images, and the Wayback Machine [3] to gather additional information on account deletion not entirely available from the videos. The analysis drew on a combination of deceptive design patterns from the ontology proposed by Gray et al. [11], which builds on many previous works, and the social network deceptive design pattern codebook from Mildner et al. [23], which has domain relevance. We also identified several new manipulative design patterns.

3 Results and Discussion

Based on iterative inductive coding, three themes emerged from significant interface changes on Instagram. While previously-referenced works generally treat the concept of manipulative design patterns as discrete, temporally-bound units of design, our analysis of the platform over its entire history unveiled a new conceptual nuance: Some manipulative design patterns in the app morphed over time in response to external barriers and pressures. The themes were: (1) provision of limited and illusory control to users in response to negative user sentiment, (2) adaptation in response to policy change, and (3) manipulative promotion of sub-apps to compete with other platforms.

3.1 Illusion of Control in Response to Negative User Sentiment

Instagram's overarching goal of increasing user engagement has driven the introduction of recommendations and enticing design features across many parts of the app. However, as engagement-enhancing strategies intensified, a portion of the user base expressed negative sentiment, feeling they had lost control over



Figure 1: This diagram outlines each of the steps taken for the empirical design study.

Table 1: This table shows the number of videos that were systematically collected and refined into more relevant lists in the first four steps of the data gathering process for the empirical design analysis.

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Total
Step 1	60	150	180	310	390	310	460	500	580	720	810	680	740	900	1010	7800
Step 2	5	31	37	41	35	40	92	112	135	89	132	69	83	108	72	1081
Step 3	5	14	24	27	22	28	66	81	95	69	112	61	73	93	51	821
Step 4	3	3	5	7	4	8	17	12	23	8	15	10	13	17	10	155

their own experience and were caught in compulsive usage patterns [12, 17]. In response, Instagram has gradually introduced several features that give users the illusion of control over their experience, but in reality are designed to offer very limited control.

In 2018, “You’re All Caught Up” was introduced (see Figure 2a). Instead of addressing the root issue of infinite scroll, Instagram introduced this feature as a stopping point. While nominally suggesting that users could pause their consumption, the design clearly does not stop the flow of content. Rather, the message is immediately followed by a new stream of suggested content, effectively perpetuating endless scrolling in a slightly disguised form. Also in 2018, the introduction of two, nearly identical settings designed to allow users to take breaks (see Figure 2b) illustrated illusory control. “Set reminder to take breaks” and “Set daily time limit” served as nominal but relatively ineffective digital self-control tools. Both settings had unique names but functioned nearly identically as easy-to-ignore reminders, failing to give significant control over the user’s experience.

In 2022, Instagram introduced the “Snooze suggested posts in feed” setting (see Figure 2c), which allowed users to hide suggested posts for 30 days. While giving users some control, its temporary nature shows that the control given to users is fleeting by design. After the 30-day period, recommended posts return, undermining any long-term empowerment to limit algorithmic suggestions. Also in 2022, a more visually subtle example of illusory control came with the (re)introduction of the chronological feed through the “Following” option (see Figure 2d). This feature, supposedly meant to offer users more control over feed content, was buried within the interface and separated from the primary feed experience. The option was placed within a drop-down menu on the top menu bar nearly imperceptibly hidden in the app’s logo, with a small arrow appearing only when you swipe up. Once selected, users can access a chronological feed but are limited since the chronological feed cannot be set as default. To access DMs, notifications, or other tabs, users must leave the chronological page and return to the algorithmically-driven feed. Overall, these cases illustrate Instagram’s evolving manipulative design patterns in response to user

sentiment by offering features that promise more control, but in their design suggest little intention to give control.

3.2 Adapting in Response to Policy Change

The process of account deletion on Instagram provides a compelling example of how manipulative design patterns evolve in response to policy changes while maintaining the platform’s objectives of retaining users. From its inception, deleting an account lacked a straightforward option. In 2010 and 2011, users had no in-app option and needed to access Instagram’s website, find Instagram’s email address, and manually email requesting account deletion, defined by the “Clinging to Accounts” manipulative design patterns [23]. By 2012, users were required to log in to the website, access account settings, and follow a lengthy series of prompts, including selecting a reason, re-entering the password, and confirming the decision multiple times, showcasing manipulative design patterns such as Clinging to Accounts and Persuasive Language [23]. In 2015, “Delete Account” was replaced by “Temporarily Deactivate Account”, and permanent account deletion was solely available (and effectively hidden) in the Help Center. This abrupt switch exemplified the “Plain Evil” [23] manipulative design pattern, leveraging trust in a familiar interface with a less meaningful replacement. Between 2021 and 2022, a “Forced Grace Period” pattern [23] was introduced, requiring users to wait one month before permanent deletion.

A critical turning point came in June of 2022 when Apple mandated that apps provide in-app account deletion [14]. For the first time, Instagram introduced a Delete Account button in its app settings. However, this did not eliminate manipulative design patterns; it merely forced them to evolve. The new process, while meeting Apple’s requirements, continued to be unnecessarily difficult. Users had to locate deletion in the convoluted settings (“Labyrinth” pattern [23]), answering many questions, including avoiding temporary deactivation to proceed with deletion (“False Hierarchy” pattern [11]). Figure 3 and Table 2 show the rising friction in steps users were required to take to delete an account. The estimation of the number of steps involved clicks/taps to navigate within and between pages, various button selections, re-entering the password,

Table 2: This table shows in more granular detail the steps involved in Instagram account deletion over the years. The number of steps was estimated by accounting for interactions in page navigation, button selection, password re-entering, search, and even email outside the platform.

Year		2010	2011	2012	2015	2021	2022		2023	2024
Device		Web	Web	Web	Web	Web	Web	iOS Only	Mobile	Mobile
Steps	Page-Related Navigation	-	2	2	6	5	5	5	7	6
	Button Selection	-	0	3	3	3	3	5	7	7
	Password Re-enter	-	-	1	1	1	1	1	1	1
	Manual Email	1	1	-	-	-	-	-	-	-
	Search	-	1	-	-	-	-	-	-	-
Total		1	4	6	10	9	9	11	15	14
Notes		No Mention of Deletion in App or Website		Hidden in Web Settings	Temporary Deactivation Introduced	1 Month Forced Grace Period		Apple Policy Change	Standardized Deletion in Accounts Center	

manual email to Instagram (outside of the platform), and search (when not findable through other forms of navigation alone). This underscores how policy changes prompted the platform to embed new forms of manipulation while outwardly complying with the policy. This theme shows the resilience of deceptive design patterns and the necessity for holistic approaches to policy and design ethics that account for such adaptive strategies.

3.3 Manipulative Promotion of Sub-Apps in Response to Competition

We use the concept of "sub-apps" to refer to parts of the platform that are distinct enough to be considered separate apps within the primary app. In some cases these sub-apps have standalone versions that users can download, while in others sub-apps simply function within the main app. We noted the following sub-apps: Stories (autoplaying images and videos, competing with Snapchat), IGTV (long-form video content, competing with YouTube), Threads (image messaging app, competing with Snapchat), Reels (short-form video, competing with TikTok), Threads (microblogging, competing with X, formerly Twitter)², Meta AI (AI chatbot, competing with ChatGPT). While all these sub-apps contain a variety of manipulative design patterns designed to increase engagement and other goals, we focus on patterns related to cross-promotion of these sub-apps. We explore five manipulative design patterns, the latter four of which are new.

3.3.1 Muscle Memory Misuse. The visual placement of a particular feature is suddenly switched to a prominent position that the user has developed muscle memory for, causing them to accidentally interact with it. This pattern has been described in the 2023 study by Kontogeorgou and colleagues [15]. Muscle Memory Misuse relies on leveraging the user's learned behavior and reflexes to use a service unintentionally. This pattern is a type of "Manipulating Choice Architecture" deceptive design pattern under the "Interface Interference" strategy [11], where the user's expectations of the

order of options shown is subverted to make a different outcome more likely than the user's goal. For a decade, the bottom menu bar remained relatively unchanged, with only aesthetic modifications to the icons. However, in 2020 and 2021, the Reels icon replaced the Explore, Post, or Notification buttons, which disrupted users' muscle memory (see Figure 4a). Another example in 2023 and 2024 was the replacement of the Share to Story button with the Post to Threads button. Users, relying on muscle memory, may have accidentally interacted with Threads rather than sharing to their Instagram Story, suggesting the misuse of user behavior to promote sub-app engagement.

3.3.2 Delayed Consent. Users begin engaging with a feature or service without initially accepting the associated terms of service or privacy policies. Later, users are interrupted with a consent prompt, requiring them to accept the terms after they've already started using the feature. This pattern is a type of "Hiding Information" deceptive design pattern under the "Sneaking" strategy [11]. It is akin to the "Hidden Costs" deceptive design pattern but in the context of terms and policies consent. We observed this pattern specifically with Meta AI. At times, users were allowed to interact with the AI chatbot without accepting the necessary terms of service upfront. However, at some point later, users would be required to explicitly consent to Meta's AI terms, leveraging the users' existing engagement to increase the likelihood of consent.

3.3.3 Forced Integration. Sub-apps are integrated into the main app with prominent placement, encouraging user interaction, but without providing users the option to turn off or hide these features. Even if users did not initially seek these sub-apps, they are placed in noticeable locations within the app, nearly forcing users to engage with them. This can be considered a manipulative design pattern under the "Forced Action" strategy [11]. Examples of "Forced Integration" within Instagram include the launch of Stories, which appeared at the top of the home feed; the IGTV buttons that persisted in the Home and Explore tabs; and Meta AI being placed in the search box. Even though most users likely did not download

²We note that the first Threads sub-app, focused on image messaging, was discontinued before the microblogging sub-app with the same name was introduced.

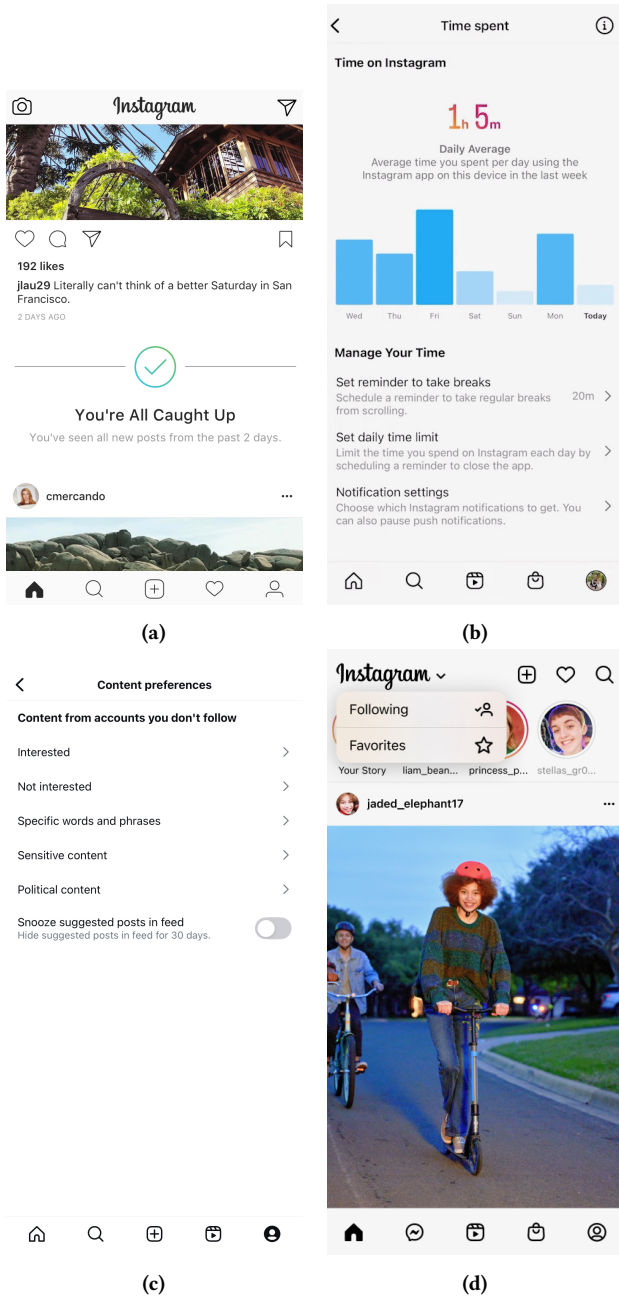


Figure 2: Examples of manipulative design patterns providing limited and illusory control over user experience on Instagram.

- (a) "You're All Caught Up" gives the illusion of a stopping point but perpetuates infinite scrolling with suggested posts underneath.
- (b) "Set daily time limit" and "Set reminder to take breaks" offer superficial control over user experience.
- (c) "Snooze suggested posts in feed" temporarily hides recommendations but reinstates them after 30 days, undermining user control.
- (d) The chronological feed is visually hidden and cannot be set as the default feed option.

Instagram to use these sub-apps, they are prominently placed without any option to hide them, ensuring that users continue to be exposed to and eventually use them within the app.

3.3.4 Tied Accounts. Multiple accounts or services are bound together in a way that limits the user's ability to manage or disengage from them independently. This pattern erodes user autonomy, prioritizing platform retention over user control. This pattern may be considered a type of "Forced Continuity" manipulative design pattern under the "Forced Action" strategy [11]. One prominent example of "Tied Accounts" is the microblogging Threads app, which required users to download a separate app. While the new app involved an account creation process, users could not delete their Threads account independently. Instead, deletion required users to delete their entire Instagram account, tying the two services together and making it difficult to disengage from one without losing access to the other.

3.3.5 Redirective Funnel. Users are subtly redirected into a secondary app or service under the guise of engaging with primary app features. This pattern is a type of "Bait and Switch" manipulative design pattern under the "Sneaking" strategy [11], where users expect to interact with one feature but are redirected to another. An example of this pattern is when users see a Reel in their feed and, upon tapping it, are redirected to the Reels sub-app. Similarly, if users click to read more in a Threads carousel, they are directed to download the Threads app (see Figure 4b). Additionally, IGTV videos, whether on the Home feed or Explore page, often cut off after a short period, prompting users to watch the video within the IGTV sub-app. These examples illustrate how Instagram funnels users into sub-apps through subtle redirection, often without users' explicit consent or intention.

Table 3 summarizes the various sub-apps and their associated manipulative design patterns, illustrating how Instagram has leveraged these strategies to drive engagement across its suite of sub-apps.

4 Limitations and Future Work

In this study, we faced several limitations. Videos published for public consumption may omit parts of the interface due to editing or selective recording, potentially leaving some areas of the interface underrepresented. Videos did not provide metadata such as app version, operating system, or country of use. This introduced ambiguity in discerning the exact context of the recorded interface, such as whether certain features are the result of A/B testing, temporary changes, or permanent updates. Moreover, variations between platforms (Android vs. iOS) were generally not considered. Also, most analyzed videos were in English, and so, the results are not globally representative. Despite these limitations, the method employed in this study offers substantial promise for analyzing historical changes in manipulative design patterns. This study underscores the utility of archival methods in analyzing manipulative design patterns. Using user-generated YouTube content as a form of "design archaeology," enabled us to capture diverse user experiences across time and capture insights that would have been difficult to obtain otherwise.

Estimated Steps to Account Deletion

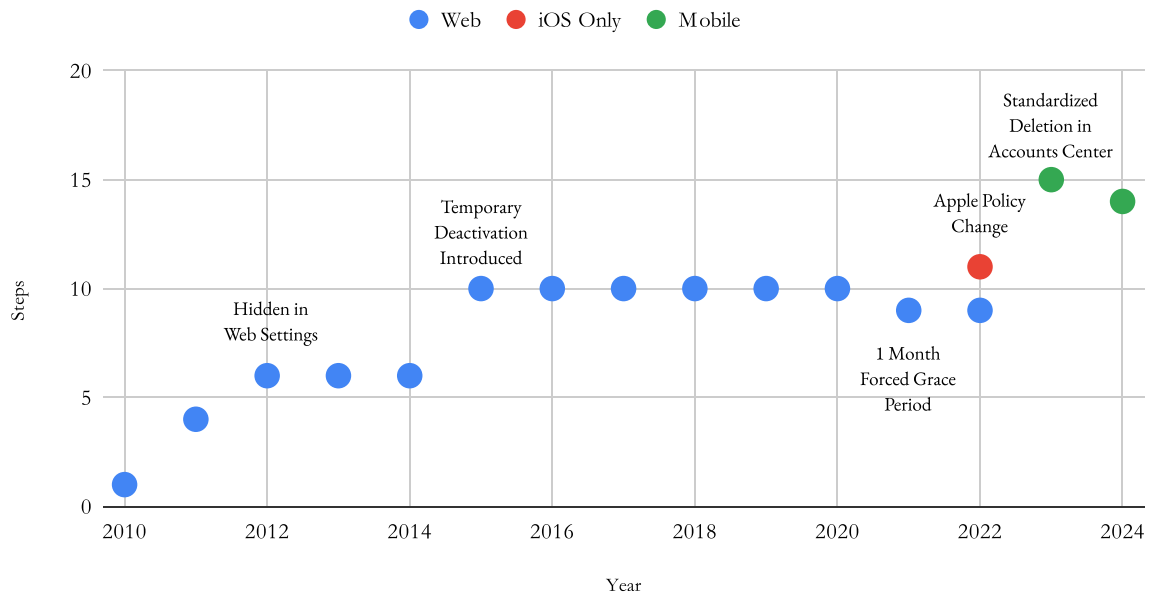


Figure 3: This graph shows an estimate of the number of steps a user had to take to delete an account over Instagram’s history. Despite a policy change from Apple in 2022 to require an in-app option for account deletion, the process has generally increased in difficulty over its history.

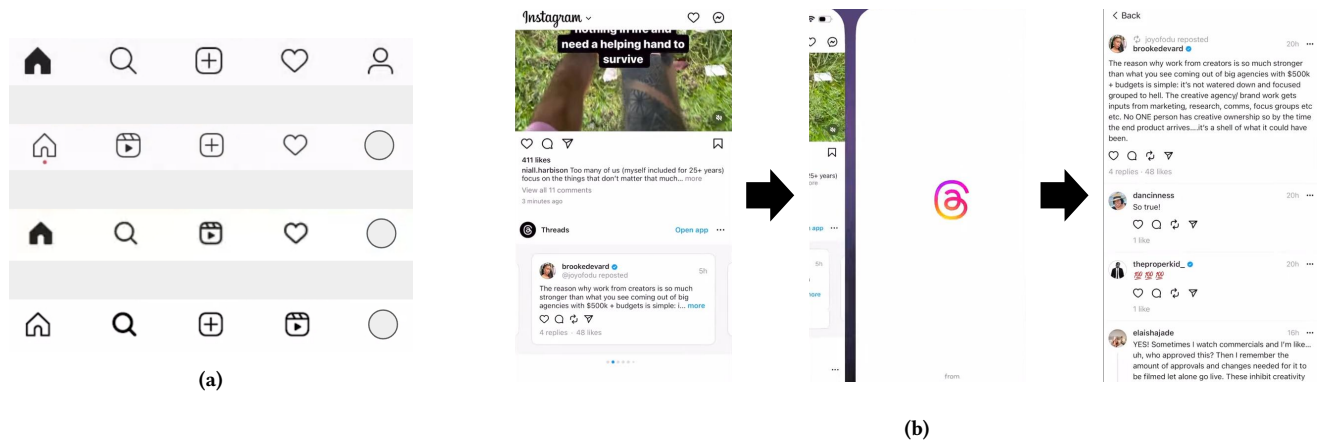


Figure 4: Two examples of manipulative design patterns related to manipulative promotion of sub-apps. (a) In Muscle Memory Misuse, the bottom menu bar of Instagram in 2020 and 2021 experienced several variations of the newly introduced Reels sub-app replacing the icons for Explore, Post, and Notifications. (b) In the Redirective Funnel, a user comes across posts from Threads within Instagram’s app and upon tapping to read more, is unexpectedly redirected to the standalone Threads app.

Future work could complement this approach by analyzing archived app versions where available, such as on websites like APKMirror [2] and OldVersion.com [27]. This would allow for a more comprehensive exploration of app features. Additionally, researchers

could develop crowd-sourced tools for documenting app interfaces, akin to the Wayback Machine [3]. Such tools could enable users to

Table 3: Table of Instagram "sub-apps" over the years with manipulative design patterns employed to steer users into using them.

Sub-Apps		Stories	IGTV	Threads	Reels	Threads	Meta AI
Year Launched		2016	2018	2019	2020	2023	2024
Main Competitor		Snapchat	YouTube	Snapchat	TikTok	X (Twitter)	ChatGPT
Format		Autoplaying images and videos	Long-form video	Image messaging	Short-form video	Microblogging	AI chatbot
Sub-App Promotion Dark Patterns	<i>Muscle Memory Misuse</i>	○	○	○	●	●	●
	<i>Delayed Consent</i>	○	○	○	○	○	●
	<i>Forced Integration</i>	●	●	○	●	○	●
	<i>Tied Accounts</i>	○	●	●	○	●	○
	<i>Redirective Funnel</i>	○	●	○	●	●	○

upload footage alongside metadata for systematically tracking interface changes across platforms over time. Extending this study to examine patterns in other applications could also provide valuable insights into how manipulative design strategies evolve in different platforms and contexts, and reveal broader industry trends. This type of analysis could also be enriched by incorporating other forms of historical data, for instance, app store reviews and user sentiment, app store policy changes, and relevant legislation.

5 Conclusion

The growing research over manipulative and deceptive design practices has highlighted their prominence across digital platforms, including social networking services. However, there have been few studies exploring how these design patterns evolve over time in social networking services. In this case study, we examined a few key themes in the evolution of deceptive design patterns in Instagram's interface over its 15-year history. The key contributions of this research include the development of an enhanced method for obtaining archival material on UI interface changes through user-generated videos available online, the introduction of an "evolution-based" thematic manipulative design patterns case study based on Instagram's mobile app interface, and the identification of four new deceptive design patterns.

Our findings uncover three key themes that illustrate how the platform adapted its manipulative design practices: (1) the provision of limited and illusory control to users in response to user sentiment, (2) adaptations to manipulative design patterns in response to policy changes, and (3) the promotion of sub-apps through manipulative design patterns in response to competition. These findings underscore the persistent and evolving nature of deceptive design patterns, suggesting that such strategies continue to adapt and evolve in response to user sentiment, regulation, and competition. Our work also suggests an urgent need for stronger regulatory frameworks and more effective tools to document and track interface design changes, enabling further research and better oversight of manipulative and deceptive design practices.

References

- [1] Jacob Aagaard, Miria Emma Clausen Knudsen, Per Bækgaard, and Kevin Doherty. 2022. A Game of Dark Patterns: Designing Healthy, Highly-Engaging Mobile Games. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI EA '22). Association for Computing Machinery, New York, NY, USA, Article 438, 8 pages. doi:10.1145/3491101.3519837
- [2] APKMirror. 2025. APKMirror. <https://www.apkmirror.com/>
- [3] Internet Archive. 2025. Wayback Machine. <https://web.archive.org/>
- [4] Harry Brignull. 2013. Dark Patterns: Inside the Interfaces Designed to Trick You. <https://www.theverge.com/2013/8/29/4640308/dark-patterns-inside-the-interfaces-designed-to-trick-you>
- [5] Deceptive Design. 2025. About Us - Deceptive Design. <https://www.deceptive.design/about-us>
- [6] Linda Di Geronimo, Larissa Braz, Enrico Fregnan, Fabio Palomba, and Alberto Bacchelli. 2020. UI Dark Patterns and Where to Find Them: A Study on Mobile Applications and User Perception. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3313831.3376600
- [7] European Data Protection Board. 2023. *Guidelines 03/2022 on Deceptive Design Patterns in Social Media Platform Interfaces: How to Recognise and Avoid Them*. Technical Report Version 2.0. https://edpb.europa.eu/system/files/2023-02/edpb_03-2022_guidelines_on_deceptive_design_patterns_in_social_media_platform_interfaces_v2_en_0.pdf
- [8] Dan Fitton and Janet C. Read. 2019. Creating a Framework to Support the Critical Consideration of Dark Design Aspects in Free-to-Play Apps. In *Proceedings of the 18th ACM International Conference on Interaction Design and Children* (Boise, ID, USA) (IDC '19). Association for Computing Machinery, New York, NY, USA, 407–418. doi:10.1145/3311927.3323136
- [9] Colin M. Gray, Yubo Kou, Bryan Battles, Joseph Hoggatt, and Austin L. Toombs. 2018. The Dark (Patterns) Side of UX Design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3173574.3174108
- [10] Colin M. Gray, Lorena Sanchez Chamorro, Ike Obi, and Ja-Nae Duane. 2023. Mapping the Landscape of Dark Patterns Scholarship: A Systematic Literature Review. In *Companion Publication of the 2023 ACM Designing Interactive Systems Conference* (Pittsburgh, PA, USA) (DIS '23 Companion). Association for Computing Machinery, New York, NY, USA, 188–193. doi:10.1145/3563703.3596635
- [11] Colin M. Gray, Cristiana Teixeira Santos, Nataliia Bielova, and Thomas Mildner. 2024. An Ontology of Dark Patterns Knowledge: Foundations, Definitions, and a Pathway for Shared Knowledge-Building. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 289, 22 pages. doi:10.1145/3613904.3642436
- [12] Alex Hern. 2018. Instagram is supposed to be friendly. So why is it making people so miserable? <https://www.theguardian.com/technology/2018/sep/17/instagram-is-supposed-to-be-friendly-so-why-is-it-making-people-so-miserable>
- [13] Shun Hidaka, Sota Kobuki, Mizuki Watanabe, and Katie Seaborn. 2023. Linguistic Dead-Ends and Alphabet Soup: Finding Dark Patterns in Japanese Apps. In

- Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 3, 13 pages. doi:10.1145/3544548.3580942
- [14] Apple Inc. 2022. Offering Account Deletion in Your App. <https://developer.apple.com/support/offering-account-deletion-in-your-app/>
 - [15] Mariliza Kontogeorgou, Christof Van Nimwegen, and Almila Akdag Salah. 2023. Illuminating Muscle Memory's Sinister Side: A Social Media Case Study. In *Proceedings of the European Conference on Cognitive Ergonomics 2023* (Swansea, United Kingdom) (ECCE '23). Association for Computing Machinery, New York, NY, USA, Article 7, 4 pages. doi:10.1145/3605655.3605664
 - [16] Monica Kowalczyk, Johanna T. Gunawan, David Choffnes, Daniel J Dubois, Woodrow Hartzog, and Christo Wilson. 2023. Understanding Dark Patterns in Home IoT Devices. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 179, 27 pages. doi:10.1145/3544548.3581432
 - [17] Taylor Lorenz. 2022. Instagram knows you don't like its changes. It doesn't care. <https://www.washingtonpost.com/technology/2022/07/27/instagram-video-shift-kardashian/>
 - [18] Kai Lukoff, Ulrik Lyngs, Himanshu Zade, J. Vera Liao, James Choi, Kaiyue Fan, Sean A. Munson, and Alexis Hiniker. 2021. How the Design of YouTube Influences User Sense of Agency. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 368, 17 pages. doi:10.1145/3411764.3445467
 - [19] Arunesh Mathur, Gunes Acar, Michael J. Friedman, Eli Lucherini, Jonathan Mayer, Marshini Chetty, and Arvind Narayanan. 2019. Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW, Article 81 (Nov. 2019), 32 pages. doi:10.1145/3359183
 - [20] Arunesh Mathur, Mihir Kshirsagar, and Jonathan Mayer. 2021. What Makes a Dark Pattern... Dark? Design Attributes, Normative Considerations, and Measurement Methods. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 360, 18 pages. doi:10.1145/3411764.3445610
 - [21] Thomas Mildner, Merle Freye, Gian-Luca Savino, Philip R. Doyle, Benjamin R. Cowan, and Rainer Malaka. 2023. Defending Against the Dark Arts: Recognising Dark Patterns in Social Media. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference* (Pittsburgh, PA, USA) (DIS '23). Association for Computing Machinery, New York, NY, USA, 2362–2374. doi:10.1145/3563657.3595964
 - [22] Thomas Mildner and Gian-Luca Savino. 2021. Ethical User Interfaces: Exploring the Effects of Dark Patterns on Facebook. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 464, 7 pages. doi:10.1145/3411763.3451659
 - [23] Thomas Mildner, Gian-Luca Savino, Philip R. Doyle, Benjamin R. Cowan, and Rainer Malaka. 2023. About Engaging and Governing Strategies: A Thematic Analysis of Dark Patterns in Social Networking Services. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 192, 15 pages. doi:10.1145/3544548.3580695
 - [24] Thomas Mildner, Gian-Luca Savino, Susanne Putze, and Rainer Malaka. 2024. Finding a Way Through the Social Media Labyrinth: Guiding Design Through User Expectations. In *Proceedings of the International Conference on Mobile and Ubiquitous Multimedia* (MUM '24). Association for Computing Machinery, New York, NY, USA, 157–171. doi:10.1145/3701571.3701605
 - [25] Alberto Monge Roffarello, Kai Lukoff, and Luigi De Russis. 2023. Defining and Identifying Attention Capture Deceptive Designs in Digital Interfaces. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 194, 19 pages. doi:10.1145/3544548.3580729
 - [26] Midas Nouwens, Ilaria Liccardi, Michael Veale, David Karger, and Lalana Kagal. 2020. Dark Patterns after the GDPR: Scraping Consent Pop-ups and Demonstrating their Influence. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376321
 - [27] OldVersion.com. 2025. OldVersion.com. <http://www.oldversion.com/android/>
 - [28] Jenny Radesky, Alexis Hiniker, Caroline McLaren, Eliz Akgun, Alexandria Schaller, Heidi M. Weeks, Scott Campbell, and Ashley N. Gearhardt. 2022. Prevalence and Characteristics of Manipulative Design in Mobile Applications Used by Children. *JAMA Network Open* 5, 6 (06 2022), e2217641–e2217641. doi:10.1001/jamanetworkopen.2022.17641
 - [29] Mohammad Hamdi Al Khasawneh Reem Rafiq Al-Tabakhi and Ala' Omar Dandis. 2024. Investigating Dark Patterns on Social Media: Implications for User Engagement and Impulse Buying Behavior. *Journal of Internet Commerce* 23, 4 (2024), 469–502. doi:10.1080/15332861.2024.2403054 arXiv:<https://doi.org/10.1080/15332861.2024.2403054>
 - [30] Lorena Sanchez Chamorro, Carine Lallemand, and Colin M. Gray. 2024. "My Mother Told Me These Things are Always Fake" - Understanding Teenagers' Experiences with Manipulative Designs. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) (DIS '24). Association for Computing Machinery, New York, NY, USA, 1469–1482. doi:10.1145/3643834.3660704
 - [31] Brennan Schaffner, Neha A. Lingareddy, and Marshini Chetty. 2022. Understanding Account Deletion and Relevant Dark Patterns on Social Media. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2, Article 417 (Nov. 2022), 43 pages. doi:10.1145/3555142
 - [32] Katie Seaborn, Tatsuya Itagaki, Mizuki Watanabe, Yijia Wang, Ping Geng, Takao Fujii, Yuto Mandai, Miu Kojima, and Suzuka Yoshida. 2024. Deceptive, Disruptive, No Big Deal: Japanese People React to Simulated Dark Commercial Patterns. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI EA '24). Association for Computing Machinery, New York, NY, USA, Article 95, 8 pages. doi:10.1145/3613905.3651099
 - [33] Carla Sousa and Ana Oliveira. 2023. The Dark Side of Fun: Understanding Dark Patterns and Literacy Needs in Early Childhood Mobile Gaming. In *Proceedings of the 17th European Conference on Games Based Learning*. Lusófona University, CICANT. doi:10.34190/ecglb.17.1.1656 Published: September 29, 2023.
 - [34] Jelmer Tiemessen, Hanna Schraffenberger, and Gunes Acar. 2023. The Time is Ticking: The Effect of Limited Time Discounts on Consumers' Buying Behavior and Experience. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 277, 11 pages. doi:10.1145/3544549.3585735